



MANIPAL

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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Cryptology – 5_Theory

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Mohammed Harris

1) Rail Fence

Encrypt the message "Enjoy your Journey" using Rail Fence with depth 2 and 3.

Depth-2

E J Y O R O R E
N O Y U J U N Y

Cipher → EJYORORE NOYUJUNY

Depth-3

E Y R R
N O Y U J U N Y
J O O E

Cipher → EYRRNOYUJUNYJOOE

2) Encrypt the message as per your choice using Rail Fence with depth 2 & 3.

Depth-2

What is it

W A I I
H T S T

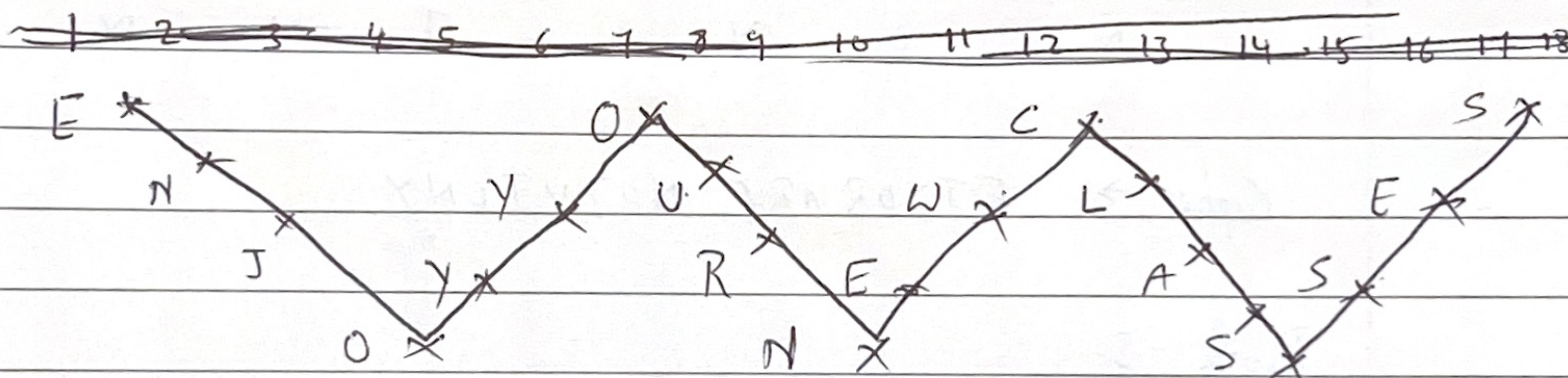
Cipher - W A I I H T S T

Depth - 3

W I
H T S T
A I

Cypher - W I H T S T A I

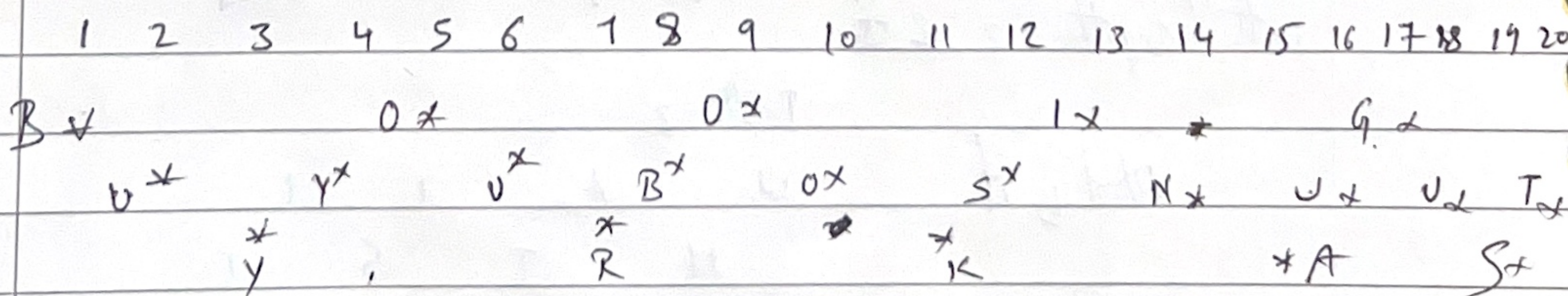
- 3) Decrypt the cypher "EOCSNYUWLEJYREASONS", encrypted using RailFence with depth 4.
D-4 length - 19



PLAIN TEXT - ENJOY your NEW CLASSES

- 4) BOOIQUYUBOSNUUTYRKAS Depth - 3

length - 20



PT - BUY YOUR BOOKS IN AUGUST

Permutation

- 1) Encrypt the following message "We are discovered" using the key "53124"

5 3 1 2 4

W E A R E

D I S C O

N E R E D

Ciphertext → ASRRCEEIEEDW
-DV

DES

- 1) Using the initial permutation table, determine the

hex → 0x0002 00000000 0001

Binary:

0000	0000	0000	0010
0000	0000	0000	0000
0000	0000	0000	0000
0000	0000	0000	0001

 input to box
(plaintext)

0000 0000 0000 0010 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 0010

Input bit 64 = 1 output position 25 = 1
Input bit 15 = 1 output pos 63 = 1
input to box
(plaintext)

64-bit output

0000 0000 0000 0000 0000 0000 1000 0000 0000 0000 0000 0000 0000 0000
0000 0000 0000 00010

Hex :-

0 x 00000008 000000002

2) Plaintext (M) - 1101 . Key (K) - 1011

Calc the resulting ciphertext using XOR.

→ PT - 1101

Key - 1011

XOR

1 1 0 1
1 0 1 1

0 1 1 0

A	B	A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

Cypher text: 0110

3) The input to S-box 1 is 000.000. what is the output.

000 000.

1st & 6th bit → row

Middle 4 bits → column

So 0th Row & 0th column, the no. in S-box → 14.

Binary conversion of 14 → 1110. / output = 1110